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MULTIPLE AGENT AUTOMATIC ROOT CAUSE ANALYSIS

Abstract

Multiple agent root cause analysis techniques are described. In an implementation, a processor performs operations to troubleshoot performance of a system operation by a plurality of domains (e.g., of a distributed system). The processor executes a system artificial intelligence (AI) agent to determine which domains of the plurality of domains include domain functions in support of the system operation, and to generate, using machine learning, queries to the determined domains based on the system operation. Domain responses are received from domain AI agents associated with the determined domains responsive to the queries and generated based on domain data associated with respective domains. A system response is generated by the system AI agent using machine learning based on the domain responses.

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Background/Summary

BACKGROUND

[0001] Distributed computing systems perform system operations by invoking different functions that are executed across multiple domains. Each domain includes a collection of computing resources (e.g., machines, systems, servers, workstations, devices, processors) to separately handle different functional dependencies of the system operations. Domain resources operate according to a common set of rules or policies that are specific to that domain; this helps organize, secure, and manage complex interactions and functional dependencies that arise from implementing the system operations. Accordingly, conventional techniques tasked with troubleshooting performance of the dependent functionality typically involves individual checks within each domain, e.g., one at a time. Multiple investigators, for instance, often collaborate as part of conventional techniques to analyze and resolve a single issue, each employing specialized knowledge and privileges within respective domains. As a result, performance of conventional techniques tasked with troubleshooting performance involve significant amounts of resources over a significant amount of time, which is inefficient with corresponding high maintenance costs and user dissatisfaction.

SUMMARY

[0002] Multiple agent root cause analysis techniques are described. In an implementation, a multi-agent service is configured to troubleshoot system operations of a distributed system, automatically and without user intervention. To do so, the service utilizes a multi-agent platform that includes domain artificial intelligence (AI) agent modules and a system AI agent module to coordinate the domain AI agent modules arrival at a goal.

[0003] Each domain AI agent module, in one or more examples, is trained or retrained via machine learning using domain data that is specific to that domain. Once trained or retrained by the domain data, each domain AI agent module is operable to generate domain responses to domain queries. A domain response indicates performance of a dependent function (i.e., a domain function) implemented by that domain in furtherance of a system operation mentioned in the domain query. In some implementations, the domain response implicates other dependent functions of the system that are implemented by other domains.

[0004] The system AI agent module, in one or more examples, is likewise trained or retrained via machine learning to respond to user queries it receives for troubleshooting issues with a system operation. In response to receipt of a user query, the system AI agent module generates one or more domain queries for receipt by the domain AI agents. Each domain query is configured to troubleshoot performance of a dependent function of the system operation as it pertains to that domain. The system AI agent is further trained or retrained via machine learning to interpret the domain responses received from the domain AI agents in reply to the domain queries to output a query response indicating one or more causes of the issues mentioned in the user queries.

[0005] This Summary introduces a selection of concepts in a simplified form that are further described below in the Detailed Description. As such, this Summary is not intended to identify essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The detailed description is described with reference to the accompanying figures. Entities represented in the figures are indicative of one or more entities and thus reference is made interchangeably to single or plural forms of the entities in the discussion.

[0007] FIG. 1 is an illustration of an environment in an example implementation that is operable to employ multiple agent automatic root cause analysis techniques described herein.

[0008] FIG. 2 depicts an example dependency tree of domain functions that support execution of a system operation performed by a distributed system as described herein.

[0009] FIG. 3 depicts a system in an example implementation showing tasks of a system agent and a plurality of domain agents that are operable to employ multiple agent automatic root cause analysis techniques as described herein.

[0010] FIG. 4 depicts a system in another example implementation showing tasks of a system agent and a plurality of domain agents that are operable to employ multiple agent automatic root cause analysis techniques as described herein.

[0011] FIG. 5 depicts a system in an example implementation showing a system agent of FIG. 1 in greater detail.

[0012] FIG. 6 depicts a system in an example implementation showing a domain agent of FIG. 1 in greater detail.

[0013] FIG. 7 illustrates an example user interface showing interactions between a client device and a system agent that is operable to employ multiple agent automatic root cause analysis techniques as described herein.

[0014] FIG. 8 is a flow diagram depicting a step-by-step procedure in an example implementation of tasks performable by a processing device for implementing multiple agent automatic root cause analysis techniques as described herein.

[0015] FIG. 9 is a flow diagram depicting a step-by-step procedure in an example implementation of tasks performable by a processing device for implementing multiple agent automatic root cause analysis techniques as described herein.

[0016] FIG. 10 illustrates an example system including various components of an example device that can be implemented as any type of computing device as described and/or utilize with reference to the previous figures to implement embodiments of the techniques described herein.

DETAILED DESCRIPTION

Overview

[0017] A distributed system is typically configured using multiple computing resources, which communicate and coordinate over a network, to execute dependent functions that support system operations. A topology of a distributed system groups logically related computing resource into several computing domains. Each domain encompasses at least one computing resource (e.g., machine, system, server, workstation, device, processor) that is configured to execute dependent functions to support execution of at least one system operation. The computing resources of each domain operate according to a common set of rules or policies that are specific to that domain. Grouping computing resources of a distributed system into different domains can facilitate organization, security, and management of complex interactions and functional dependencies that occur when system operations are implemented.

[0018] System operations performed on distributed systems often rely on multiple domain functions; a system operation may depend on multiple domain functions that execute sequentially or in parallel. Some domains are redundant to promote reliability and reduce system down time when issues arise. For example, when a first domain is not satisfying performance conditions of a distributed system, a system operations relies instead on a second domain for its domain functionality. Other domains are distinct. For example, a first domain is configurable to execute a first function to register a new profile in furtherance of a system operation. The registration function then relies on a second domain to execute a second function in furtherance of the same system operation to set parameters for initializing the profile.

[0019] A technical benefit to having individual domains within a distributed architecture is that the individual domains can be siloed from each other, thereby isolating the domains from each other, and improving security. For example, each domain is configurable to have a different set of

privileges across the system so a malicious actor with access to one domain cannot access another, restricted domain. Domain isolation also improves performance. For example, each distinct domain is configurable to further a different set of domain functions for the system operations.

[0020] While several benefits are attributable to a distributed architecture, the distributed nature of domain functions that support each system operation causes complexity in evaluating performance. Manually troubleshooting performance of a distributed system, for instance, can involve individually checking functionality of each domain (e.g., sequentially one at a time) to identify a root cause. Performance of the first domain function, for example, is evaluated separate from performance of the second domain function. In some scenarios, however, technical challenges are encountered in identifying which domain and/or which specific function to check.

[0021] An investigator with expertise at debugging some domains of the distributed system, for instance, may lack an understanding and/or access to privileges usable to investigate other domains. Therefore, in some conventional scenarios multiple investigators, each with specific knowledge of a different domain, are tasked with collaborating to identify a source of a system level problem. Debugging distributed systems in this way can be time consuming and expensive in real world scenarios. Complex and/or delayed resolutions to an operational issue in a distributed system decrease user satisfaction and increase maintenance and support costs.

[0022] Accordingly, to address these and other technical challenges associated with managing distributed systems, automated multiple agent root cause analysis techniques are described. The described techniques enable an artificial intelligence (AI) platform to provide various services (e.g., streamlined workflow automations, root cause analysis, anomaly detection, and healthy remediation) that connect multiple expert knowledge sources within a distributed system that improves operational efficiency of the system.

[0023] In an implementation, a distributed system includes multiple computing domains. Each domain of the distributed system executes different functions in furtherance of respective system operations. The distributed system also includes a plurality of artificial intelligence (AI) agents configured to collaborate towards identifying possible causes for problems experienced during execution of system operations. In general, an AI agent executes in an intelligent manner to perceive an operating environment, and autonomously perform actions that achieve goals. An AI agent may improve performance by acquiring knowledge through training, retraining, and/or machine learning of a machine-learning model.

[0024] In an implementation, at least one of the AI agents is configured as a system (or global) AI agent. The system AI agent is configurable to implement a variety of functionalities. Examples of these functionalities include an ability to execute operations to achieve a goal (e.g., to determine an answer to a user query), understand and breakdown questions, and to orchestrate tasks among other AI agents of the distributed system. Each of these other AI agents is configured as a domain (or function) AI agent in this example. The domain AI agents are registered with the system AI agent for use as domain level experts that have detailed and specialized knowledge about specific functions performed by their specific domains in furtherance of various system operations. Each domain AI agent is configured to answer queries using expert level domain knowledge and understanding of different domain states including resource states, workflow statuses, logs, metrics, alerts, and so forth.

[0025] The system AI agent executes a large language model (LLM) trained and retrained (e.g., using machine learning) to have an overall understanding about operations performed on the system. The system AI agent provides a user interface to the LLM for receiving user queries to request answers about problems experienced with system operations. The user query, for instance, supports natural language inputs that can be plainly stated in a human language to convey a high level question about the problem. The user query can also be silent about domain function dependencies, e.g., does not provide insight in what domains, if any, are involved in the potential root cause. Based on user queries, the LLM of the system AI agent automatically identifies one or

more domain functions, upon which, a problematic system operation depends. For example, the system AI agent processes the user query into one or more domain level queries.

[0026] Each domain AI agent provides a machine interface to its own LLM that is trained and retrained to employ detailed knowledge about the specific functions performed by that specific domain. At the direction of the system AI agent, the LLM of one or more of the domain AI agents automatically investigates potential causes of performance issues with the domain functions inferred from the domain level queries. For example, the machine interface of each domain AI agent receives the domain level queries generated by the system AI agent to request answers about problems experienced with system operations and/or domain functions.

[0027] In some examples, each domain agent works independently to respond to a domain query with an answer that is returned to the global agent. In this scenario, the domain AI agents are restricted from querying other domain AI agents. The system AI agent coordinates a root cause analysis of a system operation by querying the machine interface of a first domain AI agent. Based on an unsatisfactory response from the first domain AI agent, the system AI agent further coordinates the investigation by querying the machine interface of a second domain AI agent.

[0028] Based on a response from the second domain AI agent, the system AI agent generates a response to the user query.

[0029] In some implementations, the domain AI agents are configured to query other domain AI agents. For example, if a domain AI agent is incapable of generating an answer to a domain query, that domain agent may itself (or suggest the system AI agent) query another domain AI agent it has reason to believe can provide an answer, e.g., which is discerned using machine learning. The system AI agent, for instance, initiates a root cause analysis of a system operation by first querying the machine interface of a first domain AI agent. The first domain AI agent then queries the machine interface of a second domain AI agent to generate a response.

[0030] The system AI agent correlates responses received from each of the domain agents into a final response, which is then output via the user interface as a response to the user query. This coordination provided by the system AI agent preserves privileges and specialized knowledge across different domains. Different domains, for instance, may use similar terminology for different functionalities and therefore the specialized knowledge supports insights into these different uses. The system AI agent, in one or more implementations, therefore, employs general knowledge about the different domains, solely, whereas each individual domain retains control over the knowledge provided by its respective domain AI agent.

[0031] In this way, debugging distributed systems as described herein using multiple agent automated root cause analysis techniques operates faster and has increased computational efficiency than debugging a distributed system using conventional techniques. Providing answers to system problems is simplified and occurs without delay, which increases user satisfaction and decreases maintenance and support costs with increased runtime. In addition, the system AI agent user interface can be configured to output insights into the debugging process by displaying (e.g., in real time) domain queries being input to the domain AI agents, along with their corresponding responses. Providing intermediary results (e.g., feedback) obtained from the different domain AI agents improves confidence in the system AI agent solution. Further discussion of these and other examples is described in the following sections and shown using corresponding figures.

[0032] In the following discussion, an example environment is described that employs the techniques described herein. Example procedures are also described that are performable in the example environment as well as other environments. Consequently, performance of the example procedures is not limited to the example environment and the example environment is not limited to performance of the example procedures.

Example Multiple Agent Automatic Root Cause Analysis Environment

[0033] FIG. 1 is an illustration of an environment 100 in an example implementation that is operable to employ multiple agent automatic root cause analysis techniques described herein. The

illustrated environment **100** includes a distributed system **102** and a client device **104** that are communicatively coupled, one to another, via a network **106**. Computing devices, which are also referred to as computing systems, that implement the distributed system **102** and the client device **104** are configurable in a variety of ways.

[0034] A computing device, for instance, is configurable as a desktop computer, a laptop computer, a mobile device (e.g., assuming a handheld configuration such as a tablet or mobile phone), and so forth. Thus, a computing device ranges from full resource devices with substantial memory and processor resources (e.g., personal computers, game consoles) to a low resource device with limited memory and/or processing resources (e.g., mobile devices). Additionally, although a single computing device is shown, a computing device is also representative of a plurality of different devices, such as multiple servers utilized by a business to perform operations “over the cloud” as described in FIG. **10**.

[0035] The client device **104** includes a communication module **108** that is representative of functionality to communicate via the network **106** with a system manager module **110** of the distributed system **102**, e.g., as a browser, a network enabled application, and so on. The system manager module **110** is configured to implement one or more system operations **112** using hardware and software resources of the distributed system **102**, e.g., a processing device and a computer-readable storage medium. For example, the system manager module **110** executes the system operations **112** for access by the client device **104** via the network **106**, e.g., as one or more digital services.

[0036] In executing the system operations **112**, the system manager module **110** exposes the client device **104** to a plurality of domain manager modules that perform domain functions which are dependencies (i.e., dependent functions) of the system operations **112**. For example, execution of the system operations **112** depends on successful execution of one or more respective domain functions executed by domain manager modules within one or multiple domains. Examples of the domain manager modules are illustrated as domain manager module **114(1)** through domain manager module **114(N)**, which execute domain functions **116(1)** through domain functions **116(N)**. The domain manager modules **114(1)-114(N)** implement a plurality of computing domains (e.g., in a cloud network). For example, using hardware and software resources of the distributed system **102**, e.g., a processing device and a computer-readable storage medium, a domain manager module **114(1)** executes one or more domain functions **116(1)**, and a domain manager module **114(N)** to execute one or more domain functions **116(N)**. Execution of the domain functions **116(1)-116(N)** supports execution of the system operations **112**.

[0037] Imagine a cloud storage application executing on the distributed system **102**, which is made available via the network **106** to the client device **104**. During its execution, the cloud storage application executes various tasks in furtherance of providing cloud storage services. The cloud storage application invokes one or more of the system operations **112** to perform these tasks, for example, to allow users to securely access and manage their data from data stores maintained within the distributed system **102**.

[0038] The cloud storage application accesses an application program interface (API) of the system manager module **110**. From the API, the cloud storage application requests a provisioning of a data stream over the network **106** to facilitate data transfers between the client device **104** and the distributed system **102**, which allows the client device **104** to utilize the application's services. The system manager module **110** executes at least one of the system operations **112** to cause a provisioning of the data stream.

[0039] In response to executing the provisioning operation of the system operations **112**, the system manager module **110** causes an invocation of one or more of the domain functions **116(1)**. For example, the domain manager module **114(1)** may manage security of the system operations **112** to ensure privacy of data, privacy of data streams, and to prevent malicious access by unauthorized users. The provisioning operation of the system operations **112** is conditioned on successful

validation of the client device **104** and/or the user of the client device **104**. The domain manager module **114(1)** executes the domain function **116(1)** to verify that the client device **104** and/or the user have valid credentials for accessing the cloud storage application and the data maintained within the distributed system **102**. If the domain function **116(1)** fails to validate or invalidate the user of the client device **104**, the provisioning operation of the system operations **112** also fails. [0040] In further response to executing the provisioning operation of the system operations **112**, the system manager module **110** causes an invocation of the domain functions **116(N)**. For example, the domain manager module **114(N)** is responsible for implementing and managing streams provisioned on the network **106**. The system manager module **110** can execute the provisioning operation of the system operations **112** to request a new stream on the network **106**, however, the domain manager module **114(N)** executes the domain functions **116(N)** that open the stream and maintain the stream. By executing the domain functions **116(N)**, the domain manager module **114(N)** ensures reliability of streams opened in the network **106** to ensure data is exchanged seamlessly between the client device **104** and the distributed system **102** without causing interference or data corruption between two different streams. The domain manager module **114(N)** may refrain from opening a new stream, however, if the domain function **116(1)** fails. The domain manager module **114(N)** does not provision data streams on the network **106** for unauthorized actors. In this way, the provisioning operation of the system operations **112** is conditioned on successful execution of the domain functions **116(1)** and the domain functions **116(N)**.

[0041] Cloud storage services are one example of digital services implemented through execution of the system operations **112**. In some implementations, the system operations **112** support various other digital services, e.g., social media services, digital content creation services, streaming services, digital content storage services, online merchant services, and so forth. The system operations **112** aid in processing data objects, e.g., a key space, an event, a workflow, an application, a stream, a resource. As some examples, the system operations **112** enable uploads, downloads, creations, publications, exchanges, purchases, or other processes that utilize data objects of the services.

[0042] The system operations **112** depend on the domain functions **116(1)-116(N)** to perform specific acts in furtherance of the processes that utilize the data objects of the services. As some examples, the domain functions **116(1)-116(N)** enable provisioning, decommissioning, creating, modifying, configuring, encrypting, rendering, and other functions on the data objects within a particular domain of the distributed system **102**.

[0043] The system manager module **110** includes a system AI agent module **118**. The domain manager modules **114(1)-114(N)** include corresponding domain AI agent modules, which are depicted in FIG. 1 as a domain AI agent module **120(1)** of the domain manager module **114(1)**, and a domain AI agent module **120(N)** of the domain manager module **114(N)**.

[0044] The system AI agent module **118** and the domain AI agent modules **120(1)-120(N)** are implemented by respective machine learning models that execute on the computing devices of the distributed system **102** to respond to queries about performance issues with the system operations **112**. In particular, the machine learning models of the system AI agent module **118** and the domain AI agent modules **120(1)-120(N)** represent AI neural networks, which in some implementations are LLM type neural networks. For example, as individual LLMs, the system AI agent module **118** and the domain AI agent modules **120(1)-120(N)** are configured to interpret general purpose language (e.g., from text documents, images, other sources) for learning statistical relationships associated with the language during computationally intensive self-supervised and semi-supervised training processes.

[0045] The system AI agent module **118** may be trained (e.g., supervised, self-supervised) or programmed with system data **124** maintained in a system storage device **122**, e.g., a computer-readable storage media. The system data **124** (e.g., text documents, data files) relates to and supports the system operations **112** available on the distributed system **102**. The domain AI agent

modules **120(1)-120(N)** are likewise trained (e.g., supervised, self-supervised) or programmed with domain data **128(1)-128(N)** maintained in respective domain storage devices **126(1)-126(N)**. The domain data **128(1)-128(N)** (e.g., text documents) relates to respective domain functions **116(1)-116(N)** available on the distributed system **102**. For example, the domain AI agent module **120(1)** accesses domain data **128(1)** stored in the domain storage device **126(1)**, including information about the domain functions **116(1)** controlled by the domain manager module **114(1)**. The domain AI agent module **120(N)** accesses domain data **128(N)** stored in the domain storage device **126(N)**, including information about the domain functions **116(N)** controlled by the domain manager module **114(N)**.

[0046] The LLM of the system AI agent module **118** executes in the distributed system **102** to respond to queries based on the overall understanding about the system operations **112** inferred from the system data **124**. The system AI agent module **118** is configured to autonomously process received system level queries into one or more domain level queries. For example, based on the queries received by the system AI agent module **118**, the LLM of the system AI agent module **118** automatically identifies one or more of the domain functions **116(1)-116(N)** on which one or more of the system operations **112** inferred from the queries depend.

[0047] At the direction of the system AI agent module **118**, LLMs of the domain AI agent modules **120(1)-120(N)** autonomously investigate causes of performance issues with the domain functions **116(1)-116(N)** inferred from the domain level queries. Potential reasons for the issues are obtained by the LLMs of the domain AI agent modules **120(1)-120(N)** based on the detailed knowledge about the domain functions **116(1)-116(N)** determined from the domain data **128(1)-128(N)**, which are performed by that specific domain in furtherance of the system operations **112**.

[0048] As shown as displayed at the client device **104**, for instance, the system AI agent module **118** controls a user interface **130**. The user interface **130** is configured to receive an input from a user of the client device **104** in the form of a user query **132**. The user interface **130** is configured to output a response **134** to the user query **132** generated by the system AI agent module **118** in the form of an answer to the question inferred from the user query **132**. For example, the client device **104** outputs text of the user query **132**, e.g., a question from the user about problems experienced with one or more of the system operations **112**. The user query **132** can be plainly stated in a human language to convey a high level question about the problem the user is investigating. The user query **132** can also be silent about any dependencies already known to one or more of the domain functions **116(1)-116(N)**. Then, when the system AI agent module **118** generates a response **134** to the user query **132**, the client device **104** outputs text of the response **134** (e.g., an answer to the question from the user that indicates one or more problems experienced with the domain functions **116(1)-116(N)** that support the system operations **112** investigated in response to the user query **132**).

[0049] To determine the response **134** to the user query **132**, the system AI agent module **118** generates individual domain queries for receipt by one or more of the domain AI agent modules **120(1)-120(N)**. For example, each of the domain AI agent modules **120(1)-120(N)** has a machine interface configured to receive domain level queries generated by the system AI agent module **118** to request answers about problems experienced with the system operations **112** and/or the domain functions **116(1)-116(N)**. In some implementations, the system AI agent module **118** updates the user interface **130** to reflect feedback **136** about the investigation process being performed to resolve the problem identified from the user query **132**. For example, text is displayed in the user interface **130** to provide the user of the client device **104** with insight into the domain level queries sent to the domain AI agent modules **120(1)-120(N)** and/or the domain level responses received from the domain AI agent modules **120(1)-120(N)** in response to the domain level queries. The feedback **136** boosts user confidence in the accuracy of the response **134**. The response **134** enables performance of the system operations **112** for troubleshooting to a root cause attributed to one or more of the domain functions **116(1)-116(N)**.

[0050] Accordingly, debugging the distributed system **102** using multiple agent automated root cause analysis techniques as provided by the system AI agent module **118** and the domain AI agent modules **120(1)-120(N)** is less time consuming and cheaper than manually debugging the distributed system **102** using conventional techniques. Providing the response **134** to system problems identified from the user query **132** is simplified and occurs without delay (e.g., in near real time), which increases user satisfaction and decreases maintenance and support costs. In addition, the user interface **130** can be configured to output insights into the debugging process by displaying (e.g., in real time) domain queries being input to the domain AI agent modules **120(1)-120(N)**, along with their corresponding responses. Providing intermediary results (e.g., the feedback **136**) obtained from the domain AI agent modules **120(1)-120(N)** improves confidence in the solution, e.g., the response **134**.

[0051] As used herein, the term “machine-learning model” refers to a computer representation that is tunable (e.g., through training and retraining) based on inputs without being actively programmed by a user to approximate unknown functions, automatically and without user intervention. In particular, the term machine-learning model includes a model that utilizes algorithms to learn from, and make predictions on, known data by analyzing training data to learn and relearn to generate outputs that reflect patterns and attributes of the training data. Examples of machine-learning models include neural networks, convolutional neural networks (CNNs), long short-term memory (LSTM) neural networks, generative adversarial networks (GANs), decision trees, support vector machines, linear regression, logistic regression, Bayesian networks, random forest learning, dimensionality reduction algorithms, boosting algorithms, deep learning neural networks, etc.

[0052] In the illustrated example, the machine-learning model of the system AI agent module **118** and the domain AI agent modules **120(1)-120(N)** are configured using a plurality of layers having, respectively, a plurality of nodes. The plurality of layers are configurable to include an input layer, an output layer, and one or more hidden layers. Calculations are performed by the nodes within the layers via hidden states through a system of weighted connections that are “learned” during training and retraining of the machine-learning model to implement a variety of tasks.

[0053] In order to train the machine-learning model, training data (e.g., the system data **124** for the system AI agent module **118**, the domain data **128(1)** for the domain AI agent module **120(1)**, the domain data **128(N)** for the domain AI agent module **120(N)**) is received that provides examples of “what is to be learned” by that respective machine-learning model, i.e., as a basis to learn patterns from the data. The machine-learning model of the system AI agent module **118**, for instance, collects and preprocesses the system data **124** as training data that includes input features and corresponding target labels, i.e., of what is exhibited by the input features. The machine-learning model of the system AI agent module **118** then initializes parameters of its machine-learning model, which are used by its machine-learning model as internal variables to represent and process information during training and represent interferences gained through training. In an implementation, the training data for the machine-learning models described herein is separated into batches to improve processing and optimization efficiency of the parameters during training.

[0054] Training data is then received as an input by each machine-learning model and used as a basis for generating predictions based on a current state of parameters of layers and corresponding nodes, a result of which is output as output data. Output data describes an outcome of the task, e.g., as a probability of being a member of a particular class in a classification scenario.

[0055] Training of the machine-learning models described herein includes calculating a loss function to quantify a loss associated with operations performed by nodes of the machine-learning models. The calculating of the loss function, for instance, includes comparing a difference between predictions specified in the output data with target labels specified by the training data. The loss function is configurable in a variety of ways, examples of which include regret, Quadratic loss function as part of a least squares technique, and so forth.

[0056] Calculation of the loss function also includes use a backpropagation operation as part of minimizing the loss function and thereby training parameters of the machine-learning model. Minimizing the loss function, for instance, includes adjusting weights of the nodes in order to minimize the loss and thereby optimize performance of the machine-learning model in performance of a particular task. The adjustment is determined by computing a gradient of the loss function, which indicates a direction to be used in order to adjust the parameters to minimize the loss. The parameters of the machine-learning model are then updated based on the computed gradient.

[0057] This process continues over a plurality of iteration in an example until a stopping criterion is met. The stopping criterion is employed by the machine-learning models in this example to reduce overfitting of the machine-learning models, reduce computational resource consumption, and promote an ability of the machine-learning models to address previously unseen data, i.e., that is not included specifically as an example in the training data. Examples of a stopping criterion include but are not limited to a predefined number of epochs, validation loss stabilization, achievement of a performance improvement threshold, or based on performance metrics such as precision and recall.

[0058] Configuration of the training data is usable to support a variety of usage scenarios. In one example, the training data is configured as the system data **124** for the machine-learning model of the system AI agent module **118**. The system data **124** includes dependencies of the system operations **112**, historical issues encountered with the system operations **112**, and historical resolutions to the issues previously experienced with the system operations **112**. In this way, the system data **124** trains the machine-learning module of the system AI agent module **118** to efficiently predict which of the domain functions **116(1)-116(N)** is likely a cause of a performance issue encountered with one or more of the system operations **112**.

[0059] In other examples, the training data is configured as the domain data **128(1)-128(N)** for the respective machine-learning model of the domain AI agent modules **120(1)-120(N)**. For example, the domain data **128(1)** includes dependencies of the system operations **112**, dependencies of the domain functions **116(1)**, historical issues encountered with the domain functions **116(1)**, and historical resolutions to the issues previously experienced with the domain functions **116(1)**. In this way, the domain data **128(1)** trains the machine-learning module of the domain AI agent module **120(1)** to efficiently predict which of the domain functions **116(1)** and/or which of the domain functions **116(N)** is likely a cause of a performance issue encountered with one or more of the system operations **112** and/or the domain functions **116(1)**.

[0060] In general, functionality, features, and concepts described in relation to the examples above and below are employed in the context of the example procedures described in this section. Further, functionality, features, and concepts described in relation to different figures and examples in this document are interchangeable among one another and are not limited to implementation in the context of a particular figure or procedure. Moreover, blocks associated with different representative procedures and corresponding figures herein are applicable together and/or combinable in different ways. Thus, individual functionality, features, and concepts described in relation to different example environments, devices, components, figures, and procedures herein are usable in any suitable combinations and are not limited to the combinations represented by the enumerated examples in this description.

Example Multiple Agent Automatic Root Cause Analysis Techniques

[0061] The following discussion describes multiple agent automatic root cause analysis techniques that are implementable utilizing the described systems and devices. Aspects of each of the procedures are implemented in hardware, firmware, software, or a combination thereof. The procedures are shown as a set of blocks that specify operations performable by hardware and are not necessarily limited to the orders shown for performing the operations by the respective blocks. Blocks of the procedures, for instance, specify operations programmable by hardware (e.g., processor, microprocessor, controller, firmware) as instructions thereby creating a special purpose

machine for carrying out an algorithm as illustrated by the flow diagram. As a result, the instructions are storable on a computer-readable storage medium that causes the hardware to perform the algorithm. For ease of description, the techniques are described with reference back to the environment **100** of FIG. **1**.

[0062] FIG. **2** depicts an example dependency tree **200** of domain functions that support execution of an operation performed by a distributed system as described herein. An operation **112(1)** is depicted in FIG. **2** as an example of one or more of the system operations **112** depicted in FIG. **1**. The system manager module **110** executes the system operation **112(1)** within the distributed system **102**.

[0063] In this example, the execution of the system operation **112(1)** is dependent on execution of several of the domain functions **116(1)-116(N)** that are executed by the domain manager modules **114(1)-114(N)**. For example, the system operation **112(1)** is executed to enable the client device **104** to access an online merchant service implemented on the distributed system **102**. The system operation **112(1)** invokes several of the domain functions **116(1)-116(N)** to ensure the access is authorized and does not impact other services provided on the distributed system **102**.

[0064] The system operation **112(1)** is dependent on execution of the domain function **116(1)** within the domain manager module **114(1)**. For example, the domain function **116(1)** checks credentials of the client device **104** to allow or prevent access to the online merchant service.

[0065] In addition, the system operation **112(1)** is dependent on execution of a domain function **116(2)** within a domain manager module **114(2)**. For instance, the domain function **116(2)** provisions a key space within the online merchant service to enable transactions between the client device **104** and the online merchant service. This dependency of the system operation **112(1)** may be a direct dependency or indirect. For instance, execution of the system operation **112(1)** invokes the domain function **116(1)** separately from invoking the domain function **116(2)**. In other examples, execution of the system operation **112(1)** invokes the domain function **116(1)**, and the domain function **116(1)** invokes the domain function **116(2)**. In this way, execution of the system operation **112(1)** within the distributed system **102** may be dependent on execution of the domain function **116(2)** within the domain manager module **114(2)**. In other cases, execution of the system operation **112(1)** within the distributed system **102** may be dependent on execution of the domain function **116(1)** within the domain manager module **114(1)**, which is dependent on execution of the domain function **116(2)** within the domain manager module **114(2)**.

[0066] As further depicted in FIG. **2**, the system operation **112(1)** is dependent on execution of a domain function **116(3)** within a domain manager module **114(3)**. For example, the domain function **116(3)** creates a stream between the online merchant service and the client device **104** to facilitate data transfers occurring between the client device **104** and the distributed system **102** in furtherance of the online merchant service.

[0067] In addition, the system operation **112(1)** is dependent on execution of the domain function **116(N)** within the domain manager module **114(N)**. For example, the domain function **116(4)** monitors the stream opened by the domain function **116(3)** to ensure reliability and high quality of services experienced by the client device **104** when accessing the online merchant service. As described above, these dependencies of the system operation **112(1)** may be direct dependencies or indirect dependencies invoked because of the execution of the domain function **116(1)** and/or the domain function **116(2)**.

[0068] Troubleshooting performance of the system operation **112(1)** manually and/or utilizing conventional techniques can involve individually checking functionality of each of the domain manager modules **114(1)-114(N)** (e.g., one at a time) to identify a root cause. Instead, multiple agent automated root cause analysis techniques are performed by the system AI agent module **118** and the domain AI agent modules **120(1)-120(N)** to autonomously inspect performance of the domain functions **116(1)-116(N)** to which the system operation **112(1)** depends, individually.

[0069] FIG. **3** depicts a system **300** in an example implementation showing operations of a system

agent and a plurality of domain agents that are operable to employ multiple agent automatic root cause analysis techniques as described herein. The system **300** depicts operations performed for producing elements for the user interface **130**. For example, the response **134** is generated in reply to the system AI agent module **118** receiving the user query **132** from interactions between the client device **104** and the user interface **130**. Generation of the response **134** includes providing the feedback **136** to impart confidence in the response **134**.

[0070] The user query **132** pertaining to an operation **112(2)** involving performance of the distributed system **102** is received by the system AI agent module **118**. For example, the user query **132** requests information about why a request from the client device **104** to access user data maintained by a cloud storage service has failed. The LLM of the system AI agent module **118** parses text of the user query **132** to identify the system operation **112(2)** as being at least one of the system operations **112** that causes a provisioning of a data stream for the cloud storage service.

[0071] An indication of feedback **136(1)** is output in the user interface **130** to convey that the system operation **112(2)** and purpose for the user query **132** are correctly understood by the LLM of the system AI agent module **118**. The system AI agent module **118** determines that the domain function **116(1)** is performed by the domain AI agent module **120(1)** in furtherance of the system operation **112(2)** based on knowledge of the distributed system **102**, which the LLM obtains from the system data **124**.

[0072] In conjunction with the feedback **136(1)**, the system AI agent module **118** issues a domain query **302** to a machine-interface of the LLM of the domain AI agent module **120(1)**. The domain query **302** requests status, information, or other feedback about performance of the domain function **116(1)** as it relates to execution of the system operation **112(2)**. For example, the domain function **116(1)** is executed to provision a stream on the network **106**. The LLM of the domain AI agent module **120(1)** parses text of the domain query **302** to identify the domain function **116(1)** to which the system operation **112(2)** depends. Based on knowledge the LLM obtains from the domain data **128(1)**, the domain AI agent module **120(1)** outputs a domain response **304**. The domain response **304** indicates that the performance issue with the system operation **112(2)** is not caused by the domain function **116(1)**, but rather, the domain response **304** implicates a domain function **116(2)**, which is performed by a domain AI agent module **120(2)**. For example, the domain function **116(2)** is executed to verify that the client device **104** and/or the user have valid credentials for accessing the cloud storage application before the domain function **116(1)** is allowed to provision the data stream for the cloud storage service. Based on the domain response **304**, the system AI agent module **118** determines that the domain function **116(2)** is performed by the domain AI agent module **120(2)** in furtherance of the system operation **112(2)**. Another indication of feedback **136(2)** is output in the user interface **130** to convey that the domain function **116(1)** is not likely a source of the problem with the system operation **112(2)**, and that the domain function **116(2)** may be the culprit.

[0073] Along with the feedback **136(2)**, the system AI agent module **118** issues a domain query **306** to a machine-interface of the LLM of the domain AI agent module **120(2)**. The domain query **306** requests status, information, or other feedback about performance of the domain function **116(2)** as it relates to execution of the system operation **112(2)**. The LLM of the domain AI agent module **120(2)** parses text of the domain query **306** to identify the domain function **116(2)** to which the system operation **112(2)** depends. Based on knowledge the LLM obtains from domain data **128(2)**, the domain AI agent module **120(2)** outputs a domain response **308**, which suggests that the domain function **116(N)** may be a root cause of the problem with the system operation **112(2)**. For example, the domain function **116(N)** implements one of the several security checks initiated by the domain function **116(2)** to verify whether the credentials for accessing the cloud storage application are valid. The domain function **116(N)** may check encryption tokens exchanged between the client device **104** and the distributed system **102**.

[0074] Based on the domain response **308**, the system AI agent module **118** determines that the

domain function **116(N)** is performed by the domain AI agent module **120(N)** in furtherance of the system operation **112(2)**. A third indication of feedback **136(N)** is output in the user interface **130** to convey that the domain function **116(2)** is not likely a source of the problem with the system operation **112(2)**, and that the domain function **116(N)** may be the culprit.

[0075] In furtherance of outputting the feedback **136(N)**, the system AI agent module **118** issues a domain query **310** to a machine-interface of the LLM of the domain AI agent module **120(N)**. The domain query **310** requests status, information, or other feedback about performance of the domain function **116(N)** as it relates to execution of the system operation **112(2)**. The LLM of the domain AI agent module **120(N)** parses text of the domain query **310** to identify the domain function **116(N)** to which the system operation **112(2)** depends. Based on knowledge the LLM obtains from the domain data **128(N)**, the domain AI agent module **120(N)** outputs a domain response **312**, which suggests that the domain function **116(N)** is a root cause of the problem with the system operation **112(2)**.

[0076] Then, the system AI agent module **118** updates the user interface **130** to respond to the user query **132**. Using machine learning and the knowledge obtained from the system data **124** and the domain responses **304**, **308**, and **312**, the system AI agent module **118** outputs the response **134** within the user interface **130**. From the user interface **130**, the response **134** indicates an issue with the system operation **112(2)**, which is caused by the domain function **116(N)** and not the domain function **116(1)** or the domain function **116(2)**. For example, the response **134** indicates a corrupt token state exists between the client device **104** and the distributed system **102**.

[0077] FIG. 4 depicts a system **400** in another example implementation showing operations of a system agent and a plurality of domain agents that are operable to employ multiple agent automatic root cause analysis techniques as described herein. The system **400** depicts a similar construction process for producing elements for the user interface **130** as depicted in FIG. 3. For example, in response to the system AI agent module **118** receiving the user query **132** from interactions between the client device **104** and the user interface **130**, the response **134** is generated. Instead of successively issuing domain queries to the individual domain AI agent modules **120(1)**-**120(N)**, however, the system AI agent module **118** issues only one domain query, and the domain AI agent modules **120(1)**-**120(N)** autonomously issue further domain queries to other domain AI agent modules **120(1)**-**120(N)**.

[0078] The user query **132** pertaining to an operation **112(3)** involving performance of the distributed system **102** is received by the system AI agent module **118**. The LLM of the system AI agent module **118** parses text of the user query **132** to identify the system operation **112(3)**.

[0079] The feedback **136(1)** is output in the user interface **130** to convey that the system operation **112(3)** and purpose for the user query **132** are correctly understood by the LLM of the system AI agent module **118**. The system AI agent module **118** determines that the domain function **116(1)** is performed by the domain AI agent module **120(1)** in furtherance of the system operation **112(3)** based on knowledge of the distributed system **102**, which the LLM obtains from the system data **124**.

[0080] Then, the system AI agent module **118** issues a domain query **402** to the machine-interface of the LLM of the domain AI agent module **120(1)** to request information about performance of the domain function **116(1)** as it relates to execution of the system operation **112(3)**. The LLM of the domain AI agent module **120(1)** parses text of the domain query **402** and based on knowledge the LLM obtains from the domain data **128(1)**, the domain AI agent module **120(1)** outputs a domain query **404** to check if the performance issue with the system operation **112(3)** is caused by the domain function **116(N)**, and not the domain function **116(1)** as suspected by the system AI agent module **118**.

[0081] The domain query **404** is output for receipt by a machine-interface of the LLM of the domain AI agent module **120(N)**. The domain query **404** requests status, information, or other feedback about performance of the domain function **116(N)** as it relates to execution of the system

operation **112(3)**. The LLM of the domain AI agent module **120(N)** parses text of the domain query **404** to identify the domain function **116(N)** to which the system operation **112(3)** depends. Based on knowledge the LLM obtains from the domain data **128(N)**, the domain AI agent module **120(N)** outputs a domain response **406**, which suggests that the domain function **116(N)** is a root cause of the problem with the system operation **112(3)**.

[0082] Then, the system AI agent module **118** updates the user interface **130** to respond to the user query **132**. The system AI agent module **118** outputs the response **134** generated using machine learning and the knowledge obtained from the system data **124** and the domain response **406** within the user interface **130**.

[0083] FIG. 5 depicts a system **500** in an example implementation showing operation of a system agent of FIG. 1 in greater detail. The system **500** represents a system manager module, which is an example implementation of the system manager module **110**.

[0084] In addition to the elements of the system manager module **110** as depicted in FIG. 1, the system **500** includes a domain agent registrar **502**. The system **500** is configured to register each of the domain AI agent modules **120(1)-120(N)** to be associated with a different domain from the plurality of domains of the distributed system **102**. The domain agent registrar **502** enables each of the domain AI agent modules **120(1)-120(N)** to request registration with the system AI agent module **118** for assisting with troubleshooting performance of the system operations **112**. The system data **124** within the system storage device **122** maintains the domain agent registrar **502**, which indicates specific domain AI agent modules **120(1)-120(N)** within the distributed system **102** that are available to assist the LLM of the system AI agent module **118**. The system AI agent module **118** generates domain queries to the domain AI agent modules **120(1)-120(N)** within the domain agent registrar **502**, and refrains from sending domain queries to domain AI agent modules **120(1)-120(N)** that are excluded from this record. In this way, privacy and security across the domains of the distributed system **102** are protected. Those domains that have pre-registered with the system AI agent module **118** are available, solely, for implementing the user interface **130** in this example.

[0085] The system **500** also includes a record of operation dependencies **504**. The system **500** is configured to maintain the operation dependencies **504** to improve efficiency in determining which of the domain functions **116(1)-116(N)** may be implicated in a user query about a particular one of the system operations **112**. From the operation dependencies **504**, the system AI agent module **118** is configured to obtain a first listing of first available functions from a first domain including at least one first dependent operation corresponding to each of the first available functions. Likewise, the system AI agent module **118** is configured to obtain from the operation dependencies **504** a second listing of second available functions from a second domain including at least one second dependent operation corresponding to each of the second available functions, and so forth.

[0086] The system data **124** within the system storage device **122** maintains the operation dependencies **504** for quick access during execution of the system AI agent module **118**. For example, when a user query about the system operation **112(1)** is received, the LLM of the system AI agent module **118** can rely on the operation dependencies **504** listed for the system operation **112(1)** as a starting point in debugging an issue that is learned as part of training the LLM. In following the example of FIG. 2 that depicts the dependencies of the system operation **112(1)**, the operation dependencies **504** indicate that the domain function **116(1)**, the domain function **116(2)**, the domain function **116(3)**, and the domain function **116(N)** are possible causes of poor performance.

[0087] FIG. 6 depicts a system **600** in an example implementation showing a domain agent of FIG. 1 in greater detail. The system **600** represents a domain manager module, which is an example implementation of any of the domain manager modules **114(1)-114(N)**.

[0088] In addition to the elements of the domain manager modules **114(1)-114(N)** as depicted in FIG. 1, the system **600** includes domain privileges **602**. The domain privileges **602** preserve

security and protection across the multiple domains of the distributed system **102** by restricting some of the domain AI agent modules **120(1)-120(N)** from having knowledge and/or from querying others of the domain AI agent modules **120(1)-120(N)**. For example, the system **600** includes the domain privileges **602** as a set of privileges that restrict the domain AI agent module **120(N)** from accessing the domain data **128(1)**. In other cases, the system **600** includes domain privileges **602** as a set of privileges that restrict the domain AI agent module **120(1)** from accessing the domain data **128(N)**. In some examples, the domain privileges **602** include privileges for accessing other domains. For example, the system **600** includes the domain privileges **602** as set of privileges that allow the domain AI agent module **120(N)** to access the domain data **128(1)** but prevent access to the domain data **128(N)** of another one of the domain AI agent modules **120(1)-120(N)**.

[0089] The system **600** also includes function dependencies **604**. Similar to the how the operation dependencies **504** are used in the system **500**, the domain data **128(1)-128(N)** within the storage devices **126(1)-126(N)** maintain the function dependencies **604** for quick access during execution of the domain AI agent modules **120(1)-120(N)**. For example, when a domain query about the domain function **116(1)** is received, the LLM of the domain AI agent module **120(1)** can rely on the function dependencies **604** listed for the system operation **112(1)** and/or the domain function **116(1)** as a starting point in debugging an issue inferred from the user query **132**. In following the example of FIG. 4 that depicts the dependencies of the system operation **112(3)**, the function dependencies **604** indicate that the domain function **116(N)** is a possible cause of poor performance with the system operation **112(3)** and/or the domain function **116(1)**.

[0090] FIG. 7 illustrates an example user interface **700** showing interactions between a client device and a system agent that is operable to employ multiple agent automatic root cause analysis techniques as described herein. The user interface **700** is an example of the user interface **130** and is described as being displayed on the client device **104** in the context of the environment **100** depicted in FIG. 1.

[0091] In the user interface **700**, a series of graphical indications depicted in FIG. 8 as indication **702** through indication **712**. The indication **702** through the indication **712** each provide information (e.g., text) indicating one or more of the user query **132**, the response **134**, or the feedback **136** provided along the way as the system AI agent module **118** and the domain AI agent modules **120(1)-120(N)** derive the response **134**.

[0092] Within the indication **702**, the LLM of the system AI agent module **118** prompts a user of the client device **104** to convey their question about performance of one or more of the system operations **112**. The interface to the LLM of the system AI agent module **118** receives the user query **132** from interactions by the client device **104** in response to the prompt.

[0093] Within an indication **704**, the LLM of the system AI agent module **118** provides feedback to give the user of the client device **104** confidence in the LLM understanding the user query **132**, including an identity of one of the system operations **112** understood to be a subject of an investigation. For example, the system AI agent module **118** formats the user query **132** for input to the LLM of the system AI agent module **118**, which may be similar to the text of the user query **132** or may be a modification of that text to put the user query **132** into a recognizable sentence structure of the LLM.

[0094] Next, the user interface **700** provides an indication **706**. From the indication **706**, the feedback **136** obtained from investigating the issue of the user query **132** is conveyed. For example, the system AI agent module **118** outputs an identifier (e.g., a name) of the domain function **116(1)** (e.g., “function X”) and the domain function **116(N)** (e.g., “function Y”) as being dependent functions of the system operation **112(3)**.

[0095] Within the indication **706**, the system AI agent module **118** also displays results of queries submitted to the domain AI agent modules **120(1)-120(N)**. For example, the system AI agent module **118** generates a first domain query that is formatted for input to the LLM of the domain AI

agent module **120(1)**. In addition, the system AI agent module **118** formats a second domain query for input to the LLM of the domain AI agent module **120(N)**.

[0096] Within the indication **708**, the system AI agent module **118** displays the response **134** derived from the domain responses received from the domain AI agent modules **120(1)-120(N)** based on the domain queries provided to them. For example, the LLM of the system AI agent module **118** generates the response **134** to the user query **132** to indicate the issue is caused by the domain function **116(2)** and not the domain function **116(1)**. In other cases where more than one of the domain functions **116(1)-116(N)** is a source of a problem, the LLM of the system AI agent module **118** generates the response **134** to the user query **132** to indicate the issue is caused by the domain function **116(1)** and the domain function **116(2)**.

[0097] To improve future analysis performed by the system AI agent module **118** (e.g., as part of the machine learning executed by the LLM of the system AI agent module **118**), user feedback is requested within the indication **710**. The user feedback requested in the indication **710** may ask for confirmation that the response **134** is satisfactory. As depicted, the indication **710** may pose a question to ask whether the user of the client device **104** requests help fixing the issue uncovered.

[0098] Within the indication **712**, the system AI agent module **118** closes the investigation triggered by the user query **132**. For example, the system AI agent module **118** uses the feedback obtained in the indication **710** to help train the LLM of the system AI agent module **118** based on the successful (or in other cases unsuccessful) resolution.

[0099] FIG. **8** is a flow diagram depicting a step-by-step procedure **800** in an example implementation of operations performable by a processing device for implementing multiple agent automatic root cause analysis techniques as described herein. In this example, the procedure **800** starts to troubleshoot performance of a system operation by a distributed system having a plurality of domains (block **802**). For example, the user query **132** is received via the user interface **130** or the user interface **700** by the system AI agent module **118**.

[0100] Which domains of the plurality of domains include domain functions in support of the system operation is determined (block **804**). For example, the LLM of the system AI agent module **118** parses the user query **132** for an indication of one or more of the system operations **112**. In response to determining the system operations **112** associated with the user query **132**, the system AI agent module **118** checks the operation dependencies **504** and identifies one or more of the domain AI agent modules **120(1)-120(N)** that are potentially responsible for executing one or more of the domain functions **116(1)-116(N)** listed in the operation dependencies **504** to support one the system operations **112** inferred from the user query **132**.

[0101] Queries to the determined domains based on the system operation are generated (block **806**). For example, after determining that the domain function **116(N)** is a dependent function of one of the system operations **112** that is the subject of the user query **132**, the system AI agent module **118** verifies that the domain AI agent module **120(N)** appears in the domain agent registrar **502**. Upon confirming the domain AI agent module **120(N)** is registered with the system AI agent module **118**, a domain query about performance of the domain function **116(N)** is input to the machine interface of the LLM associated with the domain AI agent module **120(N)**.

[0102] Domain responses from domain AI agents associated with the determined domains are received responsive to the queries and generated based on domain data associated with respective domains (block **808**). For example, the system AI agent module **118** receives a domain response from the LLM of the domain AI agent module **120(N)**. The domain response indicates a performance issue with the domain function **116(N)**, or that there is not a performance issue with the domain function **116(N)**. In the latter case, the domain response may include a candidate domain and/or one of the domain functions **116(1)-116(N)** to be checked instead.

[0103] A system response by the system AI agent using machine learning based on the domain responses is generated (block **810**). For example, the system AI agent module **118** determines an answer to the user query **132** based on the received domain response to indicate a performance

issue with the domain function **116(N)** possibly causing the issue. In some examples, the response is output via the user interface as an answer to the user query. For example, the system AI agent module **118** formats the answer to the user query **132** to output the response **134** within the user interface **130** or the user interface **700**.

[0104] FIG. **9** is a flow diagram depicting a step-by-step procedure **900** in an example implementation of operations performable by a processing device for implementing multiple agent automatic root cause analysis techniques as described herein. To begin this example, a user query configured to troubleshoot performance of a system operation in a distributed system having a plurality of domains is received (block **902**). For example, the user query **132** is transmitted via the network **106** to the system manager module **110**. The system AI agent module **118** may format the user query **132** for input to the LLM of the system AI agent module **118**. For example, text or language of the user query **132** may be broken down by the system AI agent module **118** into a suitable question that is understandable by the LLM of the system AI agent module **118**.

[0105] A first domain involving the performance of the system operation is determined (block **904**). For example, the system AI agent module **118** analyzes the user query **132** to identify one or more of the system operations **112** as a subject of investigation.

[0106] From identifying the system operations **112**, the system AI agent module **118** determines (e.g., from the operation dependencies **504**) one or more of the domain functions **116(1)-116(N)** as possible causes of a performance issue with the system operations **112**.

[0107] In some example, this step is preceded by registering a domain AI agent from each of the computing domains. For example, a first domain AI agent and a second domain AI agent are listed in the domain agent registrar **502** as each being available domain AI agents for receiving domain queries in response to the user query **132**. The operation dependencies **504** indicate the domain functions **116(1)-116(N)** that are known by the domain AI agents. In some implementations, the operation dependencies **504** include a mapping between the domain functions **116(1)-116(N)** and the system operations **112**. The system AI agent module **118** may determine a first listing of first available functions from the first domain including at least one first dependent operation corresponding to each of the first available functions. The system AI agent module **118** may likewise determine a second listing of second available functions from the second domain including at least one second dependent operation corresponding to each of the second available functions. This facilitates the system AI agent module **118** to efficiently determine which of the domain functions **116(1)-116(N)** that are candidates as causes of an issue.

[0108] A first query for receipt by a first domain AI agent of the first domain is generated (block **906**). For example, the system AI agent module **118** generates the first domain query for display in the user interface **130** or the user interface **700** to provide the feedback **136** to the user of the client device **104** as the investigation progresses. The system AI agent module **118** may format the first domain query for input to a LLM of the domain AI agent module **120(1)**. This facilitates the domain AI agent module **120(1)** upon receipt of the domain query to efficiently determine which of the domain functions **116(1)-116(N)** that are reported as possible causes for an issue.

[0109] A first domain response from the first domain AI agent is received (block **908**). For example, the LLM of the domain AI agent module **120(1)** processes the domain query received from the system AI agent module **118** and determines from the function dependencies **604** other domains of the distributed system **102** that help with one of the system operations **112** at hand. In response to checking the domain data **128(1)** and determining that none of the domain functions **116(1)-116(N)** controlled by the domain manager module **114(1)** are experiencing problems, the domain AI agent module **120(1)** generates a domain response to the domain query for the system AI agent module **118** to learn that another domain may be implicated in the investigation. In some examples, the feedback **136** within the user interface **130** or the user interface **700** that is output for display includes an indication of the domain functions **116(1)-116(N)** checked by the domain AI agent modules **120(1)-120(N)** and its domain response.

[0110] A second query for receipt by a second domain AI agent of a second domain based on the first domain response is generated (block **910**). For example, the system AI agent module **118** generates the second domain query for display in the user interface **130** or the user interface **700** to provide the feedback **136** to the user of the client device **104** as the investigation progresses. The system AI agent module **118** may format the second domain query for input to a LLM of the domain AI agent module **120(N)**. This facilitates the domain AI agent module **120(N)** upon receipt of the domain query to efficiently determine which of the domain functions **116(1)-116(N)** that are reported as possible causes for an issue.

[0111] A second domain response is received from the second domain AI agent (block **912**). For example, the LLM of the domain AI agent module **120(N)** processes the domain query received from the system AI agent module **118** and determines from the function dependencies **604** other domains of the distributed system **102** that support one of the system operations **112** being investigated. In response to checking the domain data **128(N)** and determining that one or more of the domain functions **116(N)** controlled by the domain manager module **114(N)** are experiencing problems, the domain AI agent module **120(N)** generates a domain response to the domain query for the system AI agent module **118** to learn of the domain functions **116(N)**, which may be implicated in the investigation. In some examples, the feedback **136** within the user interface **130** or the user interface **700** that is output for display includes an indication of the domain functions **116(N)** checked by the domain AI agent module **120(N)** and its domain response.

[0112] A system response to the user query, generated by the system AI agent based on the second domain response and the first domain response is output (block **914**). For example, the system AI agent module **118** generates the response **134** to the user query **132** to indicate the issue is caused by the second function controlled by the domain manager module **114(N)** and not the first function controlled by the domain manager module **114(1)**. In other cases, two domains and their respective domain functions **116(1)-116(N)** are implicated. The system AI agent module **118** generates the response **134** to the user query **132** to indicate the issue is caused by the first function controlled by the domain manager module **114(1)** and the second function controlled by the domain manager module **114(N)**. From the response **134** displayed in the user interface **130** or the user interface **700**, the user of the client device **104** that submitted the user query **132** is able to quickly troubleshoot the specific domain functions **116(1)-116(N)** causing their issue.

Example System and Device

[0113] FIG. **10** illustrates an example system **1000** including various components of an example of one or more computing devices **1002** that can be implemented as any type of computing device as described and/or utilize with reference to the previous figures to implement embodiments of the techniques described herein. The computing devices **1002** are configurable, for example, as a server of a service provider, a device associated with a client (e.g., a client device), an on-chip system, and/or any other suitable computing device or computing system. The system **1000** is configured to provide a multi-agent service for troubleshooting the distributed system **102**, autonomously and without user intervention.

[0114] The example computing devices **1002** as illustrated include a processing system **1004**, one or more computer-readable media **1006**, and one or more input/output interfaces **1008** that are communicatively coupled, one to another. Although not shown, the computing devices **1002** further includes a system bus or other data and command transfer system that couples the various components, one to another. A system bus can include any one or combination of different bus structures, such as a memory bus or memory controller, a peripheral bus, a universal serial bus, and/or a processor or local bus that utilizes any of a variety of bus architectures. A variety of other examples are also contemplated, such as control and data lines.

[0115] The processing system **1004** is representative of functionality to perform one or more operations using hardware. Accordingly, the processing system **1004** is illustrated as including hardware elements **1010** that are configurable as processors, functional blocks, and so forth. This

includes implementation in hardware as an application specific integrated circuit or other logic device formed using one or more semiconductors. The hardware elements **1010** are not limited by the materials from which they are formed, or the processing mechanisms employed therein. For example, processors are configurable as semiconductor(s) and/or transistors (e.g., electronic integrated circuits (ICs)). In such a context, processor executable instructions are electronically executable instructions.

[0116] The computer-readable media **1006** is illustrated as including memory/storage **1012** that stores instructions that are executable to cause the processing system **1004** to perform operations. The memory/storage **1012** represents memory/storage capacity associated with one or more computer-readable media. The memory/storage **1012** includes volatile media (such as random access memory (RAM)) and/or nonvolatile media (such as read only memory (ROM), Flash memory, optical disks, magnetic disks, and so forth). The memory/storage **1012** includes fixed media (e.g., RAM, ROM, a fixed hard drive, and so on) as well as removable media (e.g., Flash memory, a removable hard drive, an optical disc, and so forth). The computer-readable media **1006** is configurable in a variety of other ways as further described below.

[0117] The input/output interfaces **1008** are representative of functionality to allow a user to enter commands and information to the computing devices **1002** and allow information to be presented to the user and/or other components or devices using various input/output devices. Examples of input devices include a keyboard, a cursor control device (e.g., a mouse), a microphone, a scanner, touch functionality (e.g., capacitive sensors, other sensors that are configured to detect physical touch), a camera (e.g., employing visible or non-visible wavelengths such as infrared frequencies to recognize movement as gestures that do not involve touch), and so forth. Examples of output devices include a display device (e.g., a monitor or projector), speakers, a printer, a network card, tactile response device, and so forth. Thus, the computing devices **1002** are configurable in a variety of ways as further described below to support user interaction.

[0118] Various techniques are described herein in the general context of software, hardware elements, or program modules. Generally, such modules include routines, programs, objects, elements, components, data structures, and so forth that perform tasks or implement particular abstract data types. The terms “module,” “functionality,” and “component” as used herein generally represent software, firmware, hardware, or a combination thereof. The features of the techniques described herein are platform independent, meaning that the techniques are configurable on a variety of commercial computing platforms having a variety of processors.

[0119] An implementation of the described modules and techniques is stored on or transmitted across some form of computer-readable media. The computer-readable media **1006** includes a variety of media that is accessed by the computing devices **1002**. By way of example, and not limitation, computer-readable media includes “computer-readable storage media” and “computer-readable signal media.”

[0120] “Computer-readable storage media” refers to media and/or devices that enable persistent and/or non-transitory storage of information (e.g., instructions are stored thereon that are executable by a processing device) in contrast to mere signal transmission, carrier waves, or signals per se. Thus, computer-readable storage media refers to non-signal bearing media. The computer-readable storage media includes hardware such as volatile and non-volatile, removable, and non-removable media and/or storage devices implemented in a method or technology suitable for storage of information such as computer readable instructions, data structures, program modules, logic elements/circuits, or other data. Examples of computer-readable storage media include but are not limited to RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, hard disks, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or other storage device, tangible media, or article of manufacture suitable to store the desired information and are accessible by a computer.

[0121] “Computer-readable signal media” refers to a signal bearing medium that is configured to

transmit instructions to the hardware of the computing devices **1002**, such as via a network. Signal media typically embodies computer readable instructions, data structures, program modules, or other data in a modulated data signal, such as carrier waves, data signals, or other transport mechanism. Signal media also include any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media include wired media such as a wired network or direct wired connection, and wireless media such as acoustic, RF, infrared, and other wireless media.

[0122] As previously described, hardware elements **1010** and computer-readable media **1006** are representative of modules, programmable device logic and/or fixed device logic implemented in a hardware form that are employed in some embodiments to implement at least some aspects of the techniques described herein, such as to perform one or more instructions. Hardware includes components of an integrated circuit or on-chip system, an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a complex programmable logic device (CPLD), and other implementations in silicon or other hardware. In this context, hardware operates as a processing device that performs program tasks defined by instructions and/or logic embodied by the hardware as well as a hardware utilized to store instructions for execution, e.g., the computer-readable storage media described previously.

[0123] Combinations of the foregoing are also be employed to implement various techniques described herein. Accordingly, software, hardware, or executable modules are implemented as one or more instructions and/or logic embodied on some form of computer-readable storage media and/or by one or more hardware elements **1010**. The computing devices **1002** are configured to implement instructions and/or functions corresponding to the software and/or hardware modules. Accordingly, implementation of a module that is executable by the computing devices **1002** as software is achieved at least partially in hardware, e.g., through use of computer-readable storage media and/or hardware elements **1010** of the processing system **1004**. The instructions and/or functions are executable/operable by one or more articles of manufacture (for example, one or more of the computing devices **1002** and/or processing system **1004**) to implement techniques, modules, and examples described herein.

[0124] The techniques described herein are supported by various configurations of the computing devices **1002** and are not limited to the specific examples of the techniques described herein. This functionality is also implementable all or in part through use of a distributed system, such as over a cloud **1014** via a platform **1016** as described below.

[0125] The cloud **1014** includes and/or is representative of a platform **1016** for resources **1018**. The platform **1016** abstracts underlying functionality of hardware (e.g., servers) and software resources of the cloud **1014**. The resources **1018** include the distributed system **102** and all its applications and/or data that can be utilized while computer processing is executed on servers that are remote from the computing devices **1002**. Resources **1018** can also include services provided by the distributed system over the network **106**, in addition to over the Internet and/or through a subscriber network, such as a cellular or wireless protocol network (e.g., Wi-Fi).

[0126] The platform **1016** abstracts resources and functions to connect the computing devices **1002** with other computing devices. The platform **1016** also serves to abstract scaling of resources to provide a corresponding level of scale to encountered demand for the resources **1018** that are implemented via the platform **1016**. Accordingly, in an interconnected device embodiment, implementation of functionality described herein is distributable throughout the system **1000**. For example, the functionality is implementable in part on the computing devices **1002** as well as via the platform **1016** that abstracts the functionality of the cloud **1014**.

[0127] In implementations, the platform **1016** employs a “machine learning model” that is configured to implement the techniques described herein. A machine learning model refers to a computer representation that can be tuned (e.g., trained and retrained) based on inputs to

approximate unknown functions. In particular, the term machine learning model can include a model that utilizes algorithms to learn from, and make predictions on, known data by analyzing training data to learn and relearn to generate outputs that reflect patterns and attributes of the training data. Examples of machine learning models include neural networks (e.g., an LLM), convolutional neural networks (CNNs), long short-term memory (LSTM) neural networks, decision trees, and so forth.

[0128] Although the invention has been described in language specific to structural features and/or methodological acts, it is to be understood that the invention defined in the appended claims is not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as example forms of implementing the claimed invention.

Claims

1. A method comprising: receiving a user query configured to troubleshoot performance of a system operation having a plurality of domains; determining, by a system artificial intelligence (AI) agent using machine learning, a first domain involving the performance of the system operation; generating, by the system AI agent, a first query for receipt by a first domain AI agent of the first domain; receiving a first domain response from the first domain AI agent; generating, by the system AI agent, a second query for receipt by a second domain AI agent of a second domain based on the first domain response; receiving a second domain response from the second domain AI agent of the second domain; and outputting a system response to the user query generated by the system AI agent based on the second domain response and the first domain response.
2. The method of claim 1, further comprising registering the first domain AI agent and the second domain AI agent to each be one of a plurality of domain AI agents available to the system AI agent for receiving domain queries.
3. The method of claim 2, wherein the registering the first domain AI agent comprises: obtaining with the system AI agent a first listing of first available domain functions from the first domain including at least one first dependent system operation corresponding to each of the first available functions; and obtaining with the system AI agent a second listing of second available domain functions from the second domain including at least one second dependent system operation corresponding to each of the second available domain functions.
4. The method of claim 1, wherein the performance of the system operation is dependent on execution of a first domain function within the first domain.
5. The method of claim 4, wherein the performance of the system operation is further dependent on execution of a second domain function within the second domain.
6. The method of claim 5, wherein execution of the first domain function is dependent on execution of the second domain function.
7. The method of claim 1, wherein: the first domain includes a first set of privileges that restrict the first domain AI agent from accessing the second domain; and the second domain includes a second set of privileges that restrict the second domain AI agent from accessing the first domain.
8. The method of claim 1, wherein the generating the first domain query further comprises generating the first domain query for display in a user interface for the system AI agent, and further comprising outputting the first domain response for display in the user interface.
9. The method of claim 8, wherein the generating the second domain query further comprises generating the second domain query for display in the user interface, and further comprising outputting the second domain response for display in the user interface.
10. The method of claim 1, further comprising generating the system response to the user query to indicate an issue is caused by the first domain and the second domain.
11. The method of claim 1, further comprising generating the system response to the user query to indicate an issue is caused by the second domain and not the first domain.

- 12.** The method of claim 1, further comprising formatting the user query for input to a large language model of the system AI agent.
- 13.** The method of claim 1, further comprising: formatting the first domain query for input to a first large language model of the first domain AI agent; and formatting the second domain query for input to second large language model of the second domain AI agent.
- 14.** The method of claim 1, wherein the plurality of domains are implemented in a cloud network.
- 15.** A system comprising: a first domain artificial intelligence (AI) agent module trained via machine learning using first domain data associated with a first domain from a plurality of computing domains of the system to generate a first domain response to a first domain query about performance of a first domain function implemented by the first computing domain in furtherance of a system operation supported by the first domain function and at least one second domain function implemented by a second domain from the plurality of computing domains; and a system AI agent module implemented by a machine learning model to generate the first domain query for receipt by the first domain AI agent to troubleshoot performance of the system operation and output a response that indicates an issue with the system operation based on the first domain response.
- 16.** The system of claim 15, further comprising a second domain artificial intelligence (AI) agent module trained via machine learning using second domain data associated with the second domain to generate a second domain response to a second domain query about performance of the second domain function in furtherance of the system operation.
- 17.** The system of claim 16, wherein the system AI agent module generates the second domain query for receipt by the second domain AI agent to troubleshoot performance of the operation, and outputs the response that indicates the issue with the system operation based on the first domain response and the second domain response.
- 18.** The system of claim 16, wherein the first domain AI agent module generates the second domain query for receipt by the second domain AI agent to troubleshoot performance of the system operation, and outputs the second domain response to the system AI agent module.
- 19.** A computing device comprising: a processor; and a computer-readable storage medium storing instructions that, responsive to execution by the processor, causes the processor to perform operations including: troubleshooting performance of a system operation having a plurality of domains, the troubleshooting including: determining, by a system artificial intelligence (AI) agent, which domains of the plurality of domains include domain functions in support of the system operation; generating, by the system AI agent using machine learning, queries to the determined domains based on the system operation; receiving domain responses from domain AI agents associated with the determined domains responsive to the queries, the domain responses generated based on domain data associated with respective domains; and generating a system response by the system AI agent using machine learning based on the domain responses.
- 20.** The computing device of claim 19, the troubleshooting further including outputting the system response for display in a user interface.
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